

IPNSDP: A Solver for Nonlinear Semidefinite Programming with Automatic Differentiation and Chordal Decomposition

A.G. Weldeyesus^{*} M.F. Anjos[†]

2025

Abstract

This paper introduces IPNSDP, a MATLAB solver for general Nonlinear Semidefinite Programming (NSDP) based on a second-order interior point method. Unlike linear SDP solvers, existing NSDP solvers are very limited and often require users to provide difficult manual derivatives or rely on other modeling interfaces. IPNSDP addresses this by directly integrating with CasADi for automatic differentiation, significantly reducing user effort. For large-scale and structured sparse problems, it supports chordal decomposition, replacing large positive semidefinite constraints with smaller ones, enhanced by a merging strategy that combines submatrices with significant overlap to improve computational efficiency. By integrating automatic differentiation with optimized decomposition, IPNSDP offers a user-friendly scalable solution for a broad class of NSDP problems.

Keywords: Nonlinear semidefinite programming, interior point methods, Chordal decomposition

1 Introduction

This paper presents IPNSDP, a MATLAB-based general-purpose solver to tackle a broad class of Nonlinear Semidefinite Programming (NSDP) problems. IPNSDP implements a second-order primal-dual interior point method and leverages automatic differentiation from CasADi to automate the computation of all required derivatives. In addition, the solver exploits chordal decomposition to efficiently handle large-scale sparse and structured matrix inequalities, significantly enhancing computational scalability.

NSDP generalizes the well-established framework of linear semidefinite programming to accommodate nonlinear objectives and constraints. NSDP arises in diverse application areas, including finance [7], control theory [10], structural optimization [12, 26], polynomial optimization [19], machine learning [11] etc. However, the non-linear nature of these problems introduces challenges, particularly in derivative evaluation, which often involves a difficult matrix calculus.

While several robust and efficient solvers exist for linear semidefinite programming, such as SDPT3 [22], SeDuMi [21], and MOSEK [16], the tools available for solving NSDPs remain relatively limited [30]. In most cases, solvers for NSDPs such as PENNON [14] and PENLAB [4] have been used. These solvers are implemented using the augmented Lagrangian method and can handle a broad class of NSDPs. However, users are required to manually implement derivatives of nonlinear functions involving matrix calculus or to rely on interfaces such as YALMIP [15].

^{*}School of Mathematics and Maxwell Institute for Mathematical Sciences The University of Edinburgh, Peter Guthrie Tait Road, Edinburgh EH9 3FD, United Kingdom. Corresponding author: a.weldeyesus@ed.ac.uk

[†]School of Mathematics and Maxwell Institute for Mathematical Sciences The University of Edinburgh, Peter Guthrie Tait Road, Edinburgh EH9 3FD, United Kingdom. GERAD, 3000, chemin de la Côte-Sainte-Catherine, Montréal, QC H3T 2A7, Canada.

IPNSDP is developed to serve as a supportive alternative for tackling NSDPs, particularly those exhibiting convexity. It combines a user-friendly MATLAB interface with CasADi [1] an automatic differentiation engine. Therefore, users are only required to define the objective and constraint functions. All relevant first- and second-order derivatives are computed internally. This substantially reduces the modeling complexity of NSDPs and simplifies the application of NSDP methods to practical problems.

Furthermore, IPNSDP employs a well-known and efficient chordal decomposition technique [5, 9, 23, 32] to transform positive semidefinite constraints on large, structured sparse matrices into multiple positive semidefinite constraints on smaller submatrices. Such structured and sparse NSDPs commonly arise in applications such as structural optimization [13, 26]. To further enhance the efficiency of this decomposition, a graph-based clique merging strategy [6] is utilized to combine submatrices with significant overlap. The overall processes of decomposition and merging are all customizable by the user through the options included in the solver. This combined approach substantially reduces the computational cost of the solution algorithm and improves the numerical tractability of many sparse and structured NSDPs.

The paper is organized as follows. Section 2 describes the general NSDP problem formulation addressed by IPNSDP. Section 3 explains the theoretical background and implementation of chordal decomposition and clique merging. The implemented primal-dual interior point method and its associated parameters are detailed in Section 5. Section 5.1 describes how to use the solver, including problem setup and customization options. Several examples from the literature are solved in Section 6. Finally, concluding remarks are provided in Section 7.

2 Problem Formulation

The IPNSDP addresses a general NSDP problem of the form

$$\begin{aligned}
& \min_{X_k \in \mathbb{S}^{d_k}, u \in \mathbb{R}^n} && f(X, u) \\
& \text{subject to} && \sum_{k=1}^{m_X} \langle A_{11,i}^{(k)}, X_k \rangle + A_{12,i} u = b_{1,i}, \quad i = 1, \dots, n_1 \\
& && \sum_{k=1}^{m_X} \langle A_{21,j}^{(k)}, X_k \rangle + A_{22,j} u \leq b_{2,j}, \quad j = 1, \dots, n_2 \\
& && c_1(X, u) = 0 \\
& && c_2(X, u) \leq 0 \\
& && \mathcal{A}_s(X, u) \succeq 0, \quad s = 1, \dots, n_{psd} \\
& && \underline{u} \leq u \leq \bar{u} \\
& && \underline{\rho}_k I_{d_k} \preceq X_k \preceq \bar{\rho}_k I_{d_k}, \quad k = 1, \dots, m_X.
\end{aligned} \tag{1}$$

The problem involves two sets of decision variables. The first one is a tuple of symmetric matrices $X = (X_1, \dots, X_{m_X})$, where each matrix X_k is in the space $\mathbb{S}^{d_k}$ and is constrained to be positive semidefinite. The constants $\underline{\rho}_k$ and $\bar{\rho}_k$ define the lower and upper bound on the eigenvalues of X_k , and I_{d_k} denotes the $d_k \times d_k$ identity matrix. The other variable is a vector $u \in \mathbb{R}^n$ with $\underline{u}$ and $\bar{u}$ its lower and upper bounds. The objective function $f : \mathbb{S} \times \mathbb{R}^n \rightarrow \mathbb{R}$ and the nonlinear constraint functions $c_1 : \mathbb{S} \times \mathbb{R}^n \rightarrow \mathbb{R}^{q_1}$ and $c_2 : \mathbb{S} \times \mathbb{R}^n \rightarrow \mathbb{R}^{q_2}$ are assumed to be sufficiently smooth. The linear constraints involve the inner product,

$$\langle A, X \rangle := \text{trace}(A^T X). \tag{2}$$

with symmetric coefficient matrices $A_{11,i}^{(k)} \in \mathbb{S}^{d_k}$ and $A_{21,j}^{(k)} \in \mathbb{S}^{d_k}$ for $i = 1, \dots, n_1$ and $j = 1, \dots, n_2$, respectively. Moreover, $A_{12,i} \in \mathbb{R}^{1 \times n}$ and $A_{22,j} \in \mathbb{R}^{1 \times n}$. The terms $b_{1,i}$ and $b_{2,j}$ are

right hand side constants. Each map $\mathcal{A}_s : \mathbb{S} \times \mathbb{R}^n \rightarrow \mathbb{S}^{t_s}$ produces a symmetric positive matrix of size $t_s \times t_s$.

Remark 1. While Problem 1 includes explicit linear constraints, it may be simpler to embed them within the nonlinear constraint function for IPNSDP. This avoids manually constructing large coefficient matrices.

Note that if we replace the objective function $f(X, u)$ with a linear function and remove the nonlinear constraints in problem (1), the formulation reduces to the well-known standard primal linear SDP of the form

$$\begin{aligned}
& \underset{X_k \in \mathbb{S}^{d_k}, u \in \mathbb{R}^n}{\text{minimize}} && \sum_{k=1}^{m_X} \langle C_k, X_k \rangle + c_u^T u \\
& \text{subject to} && \\
& && \sum_{k=1}^{m_X} \langle A_{11,i}^{(k)}, X_k \rangle + A_{12,i} u = b_{1,i}, \quad i = 1, \dots, n_1 \\
& && \sum_{k=1}^{m_X} \langle A_{21,j}^{(k)}, X_k \rangle + A_{22,j} u \leq b_{2,j}, \quad j = 1, \dots, n_2 \\
& && \underline{u} \leq u \leq \bar{u} \\
& && \underline{\rho}_k I_{d_k} \preceq X_k \preceq \bar{\rho}_k I_{d_k}, \quad k = 1, \dots, m_X.
\end{aligned} \tag{3}$$

Remark 2. Although the linear SDP problem 3 can be solved using IPNSDP, it is generally recommended to use other efficient solvers specifically designed for linear SDPs. IPNSDP is primarily intended as an alternative solver for NSDPs. Nevertheless, IPNSDP can be useful for linear (dual) SDPs that exhibit structured sparsity, which can be exploited through chordal decomposition (see Section 3), a feature integrated into IPNSDP.

3 Chordal decomposition and merging

This section provides a brief overview of the chordal decomposition and merging strategy implemented in IPNSDP. The decomposition technique transforms positive semidefinite constraints on large, structured sparse matrices into multiple positive semidefinite constraints on smaller submatrices. The merging strategy improves the effectiveness of the decomposition technique by combining submatrices with significant overlap, for example, see Table 2 for the overall efficiency of these techniques.

For a detailed theoretical discussion of chordal decomposition and merging techniques, we refer the reader to [5, 9, 23, 32] and [6].

Remark 3. This section closely follows the presentation and content of Section 3 of [26], and is included for completeness and ease of readability.

The chordal decomposition originates from graph theory, where the relevant basic theories are as follows. An undirected graph $G = (\mathcal{V}, \mathcal{E})$ is defined by a finite set of vertices $\mathcal{V} = \{1, \dots, n\}$ and a set of undirected edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$, where $(\nu_i, \nu_j) \in \mathcal{E}$ implies $(\nu_j, \nu_i) \in \mathcal{E}$. A *path* in G is a sequence of vertices $\{\nu_1, \nu_2, \dots, \nu_k\} \subseteq \mathcal{V}$ such that the consecutive edges $(\nu_1, \nu_2), (\nu_2, \nu_3), \dots, (\nu_{k-1}, \nu_k)$ all belong to $\mathcal{E}$. A path $\{\nu_1, \nu_2, \dots, \nu_k\}$, with $k \geq 3$, forms a *cycle* if $(\nu_k, \nu_1) \in \mathcal{E}$, i.e., if the sequence begins and ends at the same vertex. A *chord* in a cycle is an edge connecting two nonconsecutive vertices of the cycle. A graph is said to be *chordal* if every cycle of length at least four ($k \geq 4$) contains a chord. A *clique* $\mathcal{C} \subseteq \mathcal{V}$ is a subset of vertices such that every pair of distinct vertices in $\mathcal{C}$ is connected by an edge in $\mathcal{E}$. A clique is *maximal* if it is not strictly contained in any other clique of the graph.

To apply the chordal decomposition technique to the nonlinear semidefinite programming problem (1), the slack positive semidefinite matrix variable S_s is introduced to the nonlinear positive semidefinite inequality and it is written as

$$\mathcal{A}_s(X, u) - S_s = 0. \quad (4)$$

Hence, the matrix S_s inherits the sparsity pattern of $\mathcal{A}_s(X, u)$, and it is assumed that it will be sparse and well-structured, for example see Figure 1b. To employ chordal decomposition, the graph induced by the sparsity pattern of S_s must be chordal. In general, this sparsity pattern is typically not chordal in the first instance. In such cases, a chordal extension S_s^H is constructed by adding fill-ins (i.e., edges) to ensure chordality. Finding a minimum chordal extension is NP-complete [31], hence heuristic methods are employed. IPNSDP adopts a Cholesky factorization with zero fill-in, following the procedure described in [2], see Algorithm 1.

Algorithm 1 Chordal extension and decomposition

Input: The sparsity structure of matrix S_s . This is generated by the CasADi symbolic representation the matrix $\mathcal{A}_s(X, u)$.

Output: Maximal cliques $\mathcal{C}_1, \dots, \mathcal{C}_r$.

- 1: Construct the graph induced by the sparsity pattern of S_s .
 - 2: Compute the Laplacian A_s of the graph and update $A_s \leftarrow A_s + I$.
 - 3: Compute the Cholesky factorization $PA_sP^T = LL^T$, where P is obtained from approximate minimum degree ordering.
 - 4: Form the chordal extension $S_s^H \leftarrow P^T(L + L^T)P$.
 - 5: Extract the maximal cliques $\mathcal{C}_1, \dots, \mathcal{C}_r$ (using MATLAB code based on [3]).
-

Assume now that the chordal extension S_s^H has a chordal sparsity graph $G(\mathcal{V}, \mathcal{E})$ with (possibly merged) maximal cliques $\mathcal{C}_1, \dots, \mathcal{C}_r$. Then, the inequality in (4) can be reformulated as

$$\begin{aligned} \mathcal{A}_s(X, u) - S_s^H &= 0 \\ S_s^H &= \sum_{k=1}^r E_{\mathcal{C}_k}^T S_{s,k} E_{\mathcal{C}_k} \\ S_{s,k} &\succeq 0, \quad k = 1, \dots, r, \end{aligned} \quad (5)$$

where $S_{s,k} \in \mathbb{S}^{|\mathcal{C}_k|}$ are the smaller positive semidefinite matrices and $E_{\mathcal{C}_k}$ are the $|\mathcal{C}_k| \times n$ entry-selector matrices given by

$$(E_{\mathcal{C}_k})_{ij} = \begin{cases} 1, & \text{if } \mathcal{C}_k(i) = j \\ 0, & \text{otherwise,} \end{cases} \quad (6)$$

where $\mathcal{C}_k(i)$ is the i -th vertex. In IPNSDP implementation, S_s^H is eliminated and the problem is solved in the form

$$\begin{aligned} \mathcal{A}_s(X, u) - \sum_{k=1}^r E_{\mathcal{C}_k}^T S_{s,k} E_{\mathcal{C}_k} &= 0 \\ S_{s,k} &\succeq 0, \quad k = 1, \dots, r. \end{aligned} \quad (7)$$

In this case, the fill-ins introduced in the chordal extension S_s^H are enforced to be zero via equality constraints.

3.1 Clique Merging

The Cholesky-based chordal extension method described in Algorithm 1 produces a matrix S_s^H that decomposes into many small submatrices, often with significant overlaps. To improve efficiency, a clique merging strategy can be used to combine and produce fewer submatrices of

marginally larger sizes. IPNSDP employs a clique graph-based merging technique [6]. In this framework, for each pair of cliques, the weight is first computed as

$$w_{ij} = |\mathcal{C}_i|^3 + |\mathcal{C}_j|^3 - |\mathcal{C}_i \cup \mathcal{C}_j|^3, \quad i, j \in \{1, \dots, s\}.$$

Cliques corresponding to the largest positive weights are merged iteratively until all positive weights vanish. This process reduces the number of submatrices, for example, by approximately 50% for some of the test cases considered in [26].

3.2 Customizing Decomposition and Merging

The user has full control over the chordal decomposition and merging processes via solver settings. These options include

- Enabling or disabling decomposition.
- Enabling or disabling merging of cliques.
- Setting the minimum block size for decomposition.
- Defining the maximum density allowed in blocks for decomposition.
- Specifying the expected reduction in matrix size.

All parameters can be configured through the solver interface. For complete details, see Appendix A.

4 Primal-Dual interior point Method

In this section, we briefly outline the primal-dual interior point framework employed in IPNSDP to solve the nonlinear semidefinite programming problem (1). The framework follows standard and straightforward techniques for interior point methods (see [25, 27, 29]). The presentation and content closely follow Section 4 of [24]. The main goal of this section is to explicitly describe the parameters involved in the method and to explain how customizing these algorithmic parameters through the solver options may influence the convergence behaviour.

For clarity and notational simplicity, we write the problem in a compact form where that matrix variables are vectorized using the vectorization operator svec . Specifically, for any symmetric matrix $X_k \in \mathbb{S}^{d_k}$, we define introduce $x_k \in \mathbb{R}^{d_k(d_k+1)/2}$ such that

$$x_k = \text{svec}(X_k),$$

where

$$\text{svec}(X_k) := [X_{k,11}, \sqrt{2}X_{k,21}, \dots, \sqrt{2}X_{k,d_k1}, X_{k,22}, \sqrt{2}X_{k,32}, \dots, \sqrt{2}X_{k,d_k2}, \dots, X_{k,d_kd_k}]^T,$$

which stacks the unique elements of X_k into a vector while preserving inner products in (2). In fact, IPNSDP implicitly performs this transformation and carries out computations in the resulting space of positive semidefinite matrices. However, to facilitate usability and reduce modeling effort, while still accommodating a broader class of NSDPs, the input is specified in the original formulation (2) involving the matrix X .

Next, let $x = (x_1^T, \dots, x_{m_x}^T, u)$, $n_t = \sum_{k=1}^{m_x} \frac{d_k(d_k+1)}{2} + n$, $n_s = \sum_{s=1}^{n_{\text{psd}}} t_s + n_{bs}$ with n_{bs} the number of matrices X_k with finite upper bounds $\bar{\rho}_k$, and $n_g = n_2 + n_{bu}$ with n_{bu} the number of finite lower and upper bounds $\underline{u}, \bar{u}$ on u .

Then, this formulation (2) can be written as

$$\begin{aligned}
& \underset{x \in \mathbb{R}^{n_t}}{\text{minimize}} && f_0(x) \\
& \text{subject to} && h(x) = 0 \\
& && g(x) \leq 0 \\
& && \mathcal{A}(x) \succeq 0,
\end{aligned} \tag{8}$$

where $f_0 : \mathbb{R}^{n_t} \rightarrow \mathbb{R}$, $h : \mathbb{R}^{n_t} \rightarrow \mathbb{R}^{n_1}$, $g : \mathbb{R}^{n_t} \rightarrow \mathbb{R}^{n_2}$, and $\mathcal{A} : \mathbb{R}^{n_t} \rightarrow \mathbb{S}^{n_s}$ are twice continuously differentiable. To handle the inequality constraint $g(x) \leq 0$, a slack variable $s \in \mathbb{R}_+^{n_g}$ is introduced such that

$$g(x) + s = 0, \quad s \geq 0.$$

Next, introducing the barrier parameter μ , the barrier function is written as

$$\begin{aligned}
& \underset{x \in \mathbb{R}^{n_t}, s \in \mathbb{R}^{n_g}}{\text{minimize}} && f_0(x) - \mu \ln \det(\mathcal{A}(x)) - \mu \sum_{i=1}^{n_g} \ln(s_i) \\
& \text{subject to} && h(x) = 0, \quad g(x) + s = 0.
\end{aligned} \tag{9}$$

Then, the associated Lagrangian is given by

$$\mathcal{L}_\mu(x, s, \lambda, \nu) = f_0(x) - \mu \ln \det(\mathcal{A}(x)) - \mu \sum_{i=1}^{n_g} \ln(s_i) + \lambda^T h(x) + \nu^T (g(x) + s), \tag{10}$$

with multipliers $\lambda \in \mathbb{R}^{n_1}$, $\nu \in \mathbb{R}^{n_2}$. Defining the following

$$Z := \mu \mathcal{A}(x)^{-1}, \quad W := \mu S^{-1}, \quad S := \text{diag}(s),$$

the KKT conditions are written as

$$\nabla_x f_0(x) - \mathcal{G}(x)Z + \nabla h(x)^T \lambda + \nabla g(x)^T \nu = 0 \tag{11a}$$

$$-\mu S^{-1} e + \nu = 0 \tag{11b}$$

$$h(x) = 0 \tag{11c}$$

$$g(x) + s = 0 \tag{11d}$$

$$\mathcal{A}(x)Z - \mu I = 0 \tag{11e}$$

$$SWe - \mu e = 0, \tag{11f}$$

where $e = (1, \dots, 1)^T$ of appropriate size and $\mathcal{G}(x)Z$ is defined as

$$\mathcal{G}(x)Z = \begin{pmatrix} \left\langle \frac{\partial \mathcal{A}(x)}{\partial x_1}, Z \right\rangle \\ \vdots \\ \left\langle \frac{\partial \mathcal{A}(x)}{\partial x_{n_t}}, Z \right\rangle \end{pmatrix}.$$

To apply Newton method to the KKT conditions, symmetry is enforced by replacing $\mathcal{A}(x)Z = \mu I$ with

$$H_P(\mathcal{A}(x)Z) = \mu I$$

where

$$H_P(\mathcal{A}(x)Z) = \mu I, \quad H_P(Q) := \frac{1}{2}(PQP^{-1} + (PQP^{-1})^T).$$

The non-singular symmetric matrix P is chosen according to the Nesterov–Todd scaling [17, 18].

To compute the search directions $(\Delta x, \Delta s, \Delta \lambda, \Delta \nu, \Delta Z, \Delta w)$, the Newton method is applied to the perturbed KKT system (11). This gives

$$\begin{aligned}\nabla_{xx}^2 \mathcal{L}_\mu \Delta x - \mathcal{G}(x) \Delta Z + \nabla h(x)^T \Delta \lambda + \nabla g(x)^T \Delta \nu &= -\nabla_x \mathcal{L}_\mu \\ \nabla h(x) \Delta x &= -h(x) \\ \nabla g(x) \Delta x + \Delta s &= -g(x) - s \\ S \Delta w + W \Delta s &= \mu e - S W e \\ \mathcal{E} \Delta X + \mathcal{F} \Delta Z &= \mu I - H_P(\mathcal{A}(x)Z),\end{aligned}\tag{12}$$

where $\Delta X = \sum_{i=1}^{n_t} \Delta x_i \frac{\partial \mathcal{A}(x)}{\partial x_i}$.

Given an iterate $(x, s, \lambda, \nu, Z, w)$ and search directions $(\Delta x, \Delta s, \Delta \lambda, \Delta \nu, \Delta Z, \Delta w)$, the primal step length $\tilde{\alpha}_p$ and dual step length $\tilde{\alpha}_d$ are determined as

$$\tilde{\alpha}_p = \max \{ \alpha \in (0, 1] : s + \alpha \Delta s \geq (1 - \tau)s, w + \alpha \Delta w \geq (1 - \tau)w \} \tag{13a}$$

$$\tilde{\alpha}_d = \max \{ \alpha \in (0, 1] : \nu + \alpha \Delta \nu \geq (1 - \tau)\nu, Z + \alpha \Delta Z \succeq (1 - \tau)Z \}, \tag{13b}$$

where $\tau \in (0, 1)$ is the fraction-to-the-boundary parameter. Then, the new iterate $(x^+, s^+, \lambda^+, \nu^+, Z^+, w^+)$ becomes

$$\begin{aligned}x^+ &= x + \gamma \tilde{\alpha}_p \Delta x, & s^+ &= s + \gamma \tilde{\alpha}_p \Delta s, & w^+ &= w + \gamma \tilde{\alpha}_p \Delta w \\ \lambda^+ &= \lambda + \gamma \tilde{\alpha}_d \Delta \lambda, & \nu^+ &= \nu + \gamma \tilde{\alpha}_d \Delta \nu, & Z^+ &= Z + \gamma \tilde{\alpha}_d \Delta Z,\end{aligned}$$

where $\gamma \in (0, 1)$ is a given constant.

The overall primal-dual interior point method is summarized in Algorithm 2. Given prescribed optimality tolerance $\epsilon^0 > 0$, feasibility tolerance $\epsilon^f > 0$, and maximum iteration count $k_{\max}$, the algorithm employs two nested stopping criteria.

The first criterion applies to the inner loop, i.e., for a fixed barrier parameter μ_k . The inner loop terminates when

$$\left(\epsilon_\mu^0 < \max \left(\frac{\sigma}{0.1} \mu_k, \epsilon^0 - \mu_k \right) \quad \text{and} \quad \epsilon_\mu^f < \max \left(\frac{\sigma}{0.1} \mu_k, \epsilon^f \right) \right) \quad \text{or} \quad i > k_{\max}^{\text{inner}}. \tag{14}$$

The maximum inner iteration count $k_{\max}^{\text{inner}}$ is set to 3 for the first three outer iterations, corresponding to large barrier parameter values where high accuracy is not required, otherwise, it is set to 10. Here, ϵ_μ^0 and ϵ_μ^f denote the optimality and feasibility residuals, respectively, computed as the $\|\cdot\|_\infty$ -norm of the KKT conditions in (11).

The outer loop terminates when

$$\left(\epsilon_\mu^0 < \epsilon^0 \quad \text{and} \quad \epsilon_\mu^f < \epsilon^f \right) \quad \text{or} \quad \mu < \mu_{\min} \quad \text{or} \quad k_{\text{total}} > k_{\max}. \tag{15}$$

Note that if the outer iteration stops due to one of the last two conditions in (15), it indicates that the problem was not solved to the desired optimality, may be infeasible, or the initial point may have been infeasible.

Remark 4. IPNSDP allows penalization of constraints in the objective function to enhance numerical stability when desired by the user. For example, nonlinear inequalities in (8) can be penalized element-wise using $\rho \max\{0, g_i(x)\}$ or quadratic penalties. Regardless, the original constraints remain explicitly enforced. The penalization process can be customized via solver options.

Algorithm 2 Outline of the Primal-Dual interior point Method

Input Initial feasible point $(x^0, s^0, \lambda^0, \nu^0, Z^0, w^0)$ with $s^0 > 0, Z^0 \succ 0, w^0 > 0$, optimality tolerance $\epsilon^0 > 0$, feasibility tolerance $\epsilon^f > 0$, maximum iteration count $k_{\max}$, fraction-to-the-boundary parameter $\tau \in (0, 1)$, centrality parameter $\sigma \in (0, 1)$, and step scaling parameter $\gamma \in (0, 1)$.

Output Solution $(x, s, \lambda, \nu, Z, w)$.

1: Compute initial barrier parameter

$$\mu_0 = \frac{1}{n} \langle \mathcal{A}(x^0), Z^0 \rangle + \frac{1}{n_2} \langle s^0, w^0 \rangle$$

2: Set iteration counter $k \leftarrow 0$

3: **while** outer stopping criterion (15) not satisfied **do**

4: Set inner iteration counter $i \leftarrow 0$

5: **while** inner stopping criterion (14) not satisfied for $\mu = \mu_k$ **do**

6: Solve the Newton system (12) to obtain search directions

$$(\Delta x, \Delta s, \Delta \lambda, \Delta \nu, \Delta Z, \Delta w)$$

7: Compute step lengths $\tilde{\alpha}_p, \tilde{\alpha}_d \in (0, 1]$ as in (13)

8: Update variables

$$x \leftarrow x + \gamma \tilde{\alpha}_p \Delta x, \quad s \leftarrow s + \gamma \tilde{\alpha}_p \Delta s, \quad w \leftarrow w + \gamma \tilde{\alpha}_p \Delta w$$

$$\lambda \leftarrow \lambda + \gamma \tilde{\alpha}_d \Delta \lambda, \quad \nu \leftarrow \nu + \gamma \tilde{\alpha}_d \Delta \nu, \quad Z \leftarrow Z + \gamma \tilde{\alpha}_d \Delta Z$$

9: $i \leftarrow i + 1$

10: **end while**

11: Update barrier parameter

$$\mu_{k+i} = \sigma \cdot \left(\frac{1}{n} \langle \mathcal{A}(x), Z \rangle + \frac{1}{n_2} \langle s, w \rangle \right)$$

12: $k \leftarrow k + i$

13: **end while**

5 Usage and Problem Setup

5.1 Usage

To set up IPNSDP, first install the CasADi framework from <https://web.casadi.org/get/>. Next, download IPNSDP from <https://github.com/AlemsegedWeldeyesus/IPNSDP> and run the provided `install.m` script by executing `install()`. This adds the base directory and all subfolders to the MATLAB path. Finally, verify the installation by running `ipnsdp_test()`. If the test runs successfully, IPNSDP is ready for use.

Remark 5. When writing functions for the objective or nonlinear constraints, users should be aware that CasADi may require the use of `SX` or `MX` data types. This is important when the implementation involves operations such as `for`-loops, since standard MATLAB data types are not compatible with CasADi framework.

5.2 Problem Setup

To facilitate problem formulation, a structured MATLAB template, `ipnsdp_user_template.m`, provided with the IPNSDP solver and reproduced in Appendix A, serves as a basis for modeling

a wide range of nonlinear semidefinite programming (SDP) problems.

5.2.1 Minimal Required Input

IPNSDP receives input for the formulation in (1). To specify the coefficient matrices for the linear constraints, for a given matrix $A \in \mathbb{R}^{n \times n}$, define the vector $\text{vec}(A) \in \mathbb{R}^{n^2}$ as

$$\text{vec}(A) := [a_{11}, a_{21}, \dots, a_{n1}, a_{12}, a_{22}, \dots, a_{n2}, \dots, a_{1n}, a_{2n}, \dots, a_{nn}]^T,$$

which is formed by stacking the columns of A into a single vector.

Now, following formulation (1), consider

$$\text{vec}(X) = [\text{vec}(X_1)^\top \quad \text{vec}(X_2)^\top \quad \dots \quad \text{vec}(X_{m_X})^\top]^\top, \quad (16)$$

$$A_{11} = \begin{bmatrix} \text{vec}(A_{11,1}^{(1)})^\top & \dots & \text{vec}(A_{11,1}^{(m_X)})^\top \\ \text{vec}(A_{11,2}^{(1)})^\top & \dots & \text{vec}(A_{11,2}^{(m_X)})^\top \\ \vdots & \ddots & \vdots \\ \text{vec}(A_{11,n_1}^{(1)})^\top & \dots & \text{vec}(A_{11,n_1}^{(m_X)})^\top \end{bmatrix}, \quad A_{12} = \begin{bmatrix} A_{12,1}^\top \\ A_{12,2}^\top \\ \vdots \\ A_{12,n_1}^\top \end{bmatrix}, \quad (17)$$

$$A_{21} = \begin{bmatrix} \text{vec}(A_{21,1}^{(1)})^\top & \dots & \text{vec}(A_{21,1}^{(m_X)})^\top \\ \text{vec}(A_{21,2}^{(1)})^\top & \dots & \text{vec}(A_{21,2}^{(m_X)})^\top \\ \vdots & \ddots & \vdots \\ \text{vec}(A_{21,n_2}^{(1)})^\top & \dots & \text{vec}(A_{21,n_2}^{(m_X)})^\top \end{bmatrix}, \quad A_{22} = \begin{bmatrix} A_{22,1}^\top \\ A_{22,2}^\top \\ \vdots \\ A_{22,n_2}^\top \end{bmatrix}. \quad (18)$$

To solve NSDP problems using IPNSDP, the user must define at least the following fields in a structure `prob`.

- `prob.name`: (string) Problem name. (Optional)
 - `prob.problem_data`: User-defined struct for passing data to other functions. (Optional)
- Matrix variables

- `prob.nX`: m_x Number of matrix variables $X_1, \dots, X_{m_X}$. Set to 0 if none.
- `prob.dimX`: $[d_1, \dots, d_k]$ Vector of dimensions of $X_1, \dots, X_{m_X}$. Set to `[]` if none.
- `prob.lbX`: $[\underline{\rho}_1, \dots, \underline{\rho}_k]$ Vector of lower bounds for constraints $\underline{\rho}_k I \preceq X_k$. Set to `[]` if all unbounded. Use `-Inf` for selectively unbounded entries. Defaults used if not provided.
- `prob.ubX`: $[\bar{\rho}_1, \dots, \bar{\rho}_k]$ Vector of upper bounds for constraints $X_k \preceq \bar{\rho}_k I$. Set to `[]` if all unbounded. Use `Inf` for selectively unbounded entries. Defaults used if not provided.

Scalar variables

- `prob.nu`: n Number of scalar variables $u \in \mathbb{R}^n$. Set to 0 if none.
- `prob.lbu`: $\underline{u}$ Vector of lower bounds for constraints $\underline{u} \leq u$. Set to `[]` if all unbounded. Use `-Inf` for selectively unbounded entries. Defaults used if not provided. Defaults used if not provided.

- **prob.ubu:** $\bar{u}$ Vector of upper bounds for constraints $u \leq \bar{u}$. Set to [] if all unbounded. Use **Inf** for selectively unbounded entries. Defaults used if not provided. Defaults used if not provided.

PSD constraints

- **prob.nPSDcon:** n_{psd} Number of nonlinear matrix inequality constraints, i.e., $\mathcal{A}_s(X, u) \succeq 0$. Set to 0 if none.

Linear constraints

- **prob.A11:** Coefficient matrix for $\text{vec}(X)$ as in (17). Set to [] or ignore if none.
- **prob.A12:** Coefficient matrix for u as in (17). Set to [] or ignore if none.
- **prob.A21:** Coefficient matrix for $\text{vec}(X)$ as in (18). Set to [] or ignore if none.
- **prob.A22:** Coefficient matrix for u as in (18). Set to [] or ignore if none.

Function handles

- **prob.obj:** Function handle for the objective, typically $\Theta(x)$ **objective(x, prob)**.
- **prob.nlcon:** Function handle for user-defined nonlinear constraints returning three outputs [**eq**, **ineq**, **psd**], where
 - **eq:** Vector of nonlinear equality constraints. Set to [] if none.
 - **ineq:** Vector of nonlinear inequality constraints. Set to [] if none.
 - **psd:** Cell array of symmetric matrices subject to positive semidefinite constraints (e.g., $\text{psd}\{1\} = \mathcal{A}_1(X, u)$). Set to {} if none.

Initial point (optional)

- **prob.x0.X:** Cell array of initial points for $X_1, \dots, X_{m_X}$ (e.g., $\text{x0.X}\{1\} = \text{eye}(d_1)$). Defaults are used if not provided.
- **prob.x0.u:** Vector of initial points for scalar variables u . Defaults used if not provided.

See also Example 1 for an illustration of this setup applied to the nearest correlation matrix problem.

5.2.2 Output

IPNSDP returns two output, **x** and **info**

$$[\mathbf{x}, \mathbf{info}] = \text{ipnsdp_solve}(\text{prob}).$$

Here, **x** contains all solution variables, including associated slack and dual variables, while **info** is a struct providing diagnostic information such as iteration history, residual norms, feasibility flags, and timing data.

5.2.3 Solver Options

Many features of the interior point method implemented in IPNSDP can be customized through the `prob.options` field. For example,

- Chordal decomposition: Can be enabled or disabled via `chord_decomp`, with clique merging controlled by `clique_merge`.
- Constraint redundancy checks: Can be enabled or disabled through `linconCheck` and `nonlinconCheck`.
- Algorithmic parameters: Setting tolerances (`opt_tol`, `feas_tol`), centrality and fraction-to-the-boundary parameters (`sigma`, `tau`), and maximum iterations (`max_iter`).

For details on these and other options, such as the choice of solvers for the Newton system, step length strategies, and penalization schemes, see Appendix A.

6 Numerical Examples

This section presents a selection of nonlinear semidefinite programming that arise from practical applications and can be solved using IPNSDP. The complete code for these examples is included in the software distribution. All examples were solved on a PC equipped with a 13th Gen Intel(R) Core(TM) i5-1345U CPU running at 1.60 GHz with 32 GB of RAM, using an implementation in MATLAB R2023a. The algorithmic parameters for the interior point method were set to their default values, which are explicitly listed in Appendix A. The computational times reported in this table do not include preprocessing or CasADi problem setup times.

Example 1. This example presents the problem known as the scaled nearest correlation matrix problem, which arises in financial applications [7]. It seeks a correlation matrix with a prescribed condition number that best approximates a given symmetric matrix $H \in \mathbb{S}^n$. The problem has been solved in [4] and is formulated as a nonlinear semidefinite programming of the form

$$\begin{aligned} \min_{X, z} \quad & \|zX - H\|_F^2 \\ \text{s.t.} \quad & \text{diag}(zX) = \mathbf{1} \\ & I \preceq X \preceq \kappa I, \end{aligned} \tag{19}$$

where $X \in \mathbb{S}^n$, $z \in \mathbb{R}$, and $\kappa > 1$ bounds the condition number of X . This problem exhibits nonlinearity in both the objective function and the constraints.

Given H and κ , the problem can be solved using IPNSDP with the following minimal input.

```

% Problem dimensions
prob.nX      = 1;           % One matrix variable X
prob.dimX    = [n];        % X in S^n
prob.nu      = 1;           % One scalar variable z
prob.lbX     = [1];        % Enforce X >= I
prob.ubX     = [kappa];    % Enforce X <= kappa*I
prob.nPSDcon = 0;          % no PSD constraints of type A(X,u) >= 0

% Objective function
prob.obj = @(x) norm(x.u * x.X{1} - H, 'fro')^2;

% Constraints: [eq, ineq, psd]
prob.nlcon = @(x) deal( ...
    diag(x.u * x.X{1}) - 1, ... % diag(z*X) = 1
    [], ...                     % no inequalities
    {});                        % no PSD constraints

% Solve
[x, info] = ipnsdp_solve(prob);

```

For the matrix H and dimension $n = 6$ taken from [4], the code is provided in `ncm_c_b.m`, which also illustrates the consistency of the results. For other randomly generated instances of H and n , the code is available in `ncm_c.m`, and the corresponding computational statistics are presented in Table 1. In these instances, the matrix $H \in \mathbb{S}^n$, $n = 10, 20, 40, 80$ is constructed by adding symmetric Gaussian noise to a valid correlation matrix, followed by setting its diagonal entries to 1. The parameter κ is set to 10 in all cases.

Example 2. This example presents a problem arising in structural optimization. It involves optimizing both the topology and geometry of truss structures subject to global stability constraints. The design variables include the cross-sectional areas of potential bars and the coordinates of the joints shown in Figure 1a. This problem demonstrates not only the capability of IPNSDP to solve nonlinear semidefinite programming but also the effectiveness of its built-in chordal decomposition technique, which improves the scalability of problems with structured matrix inequality constraints, such as the one shown in Figure 1b. Mathematically, the problem is formulated as a (highly) nonlinear semidefinite programming as follows, assuming there are m potential bars.

$$\begin{aligned}
& \underset{a, v, u}{\text{minimize}} && l(v)^T a \\
& \text{subject to} && f^T u_j \leq \zeta \\
& && K(a, v)u = f \\
& && K(a, v) + \tau G(a, v, u) \succeq 0 \\
& && v \in \mathcal{V} \\
& && a \geq 0,
\end{aligned} \tag{20}$$

where

- a is a decision variable for the cross-sectional areas.

Table 1: Computational results for Example 1.

Matrix size n	IPM iter	CPU (s)
10	20	<1
20	21	<1
40	22	11
80	25	610

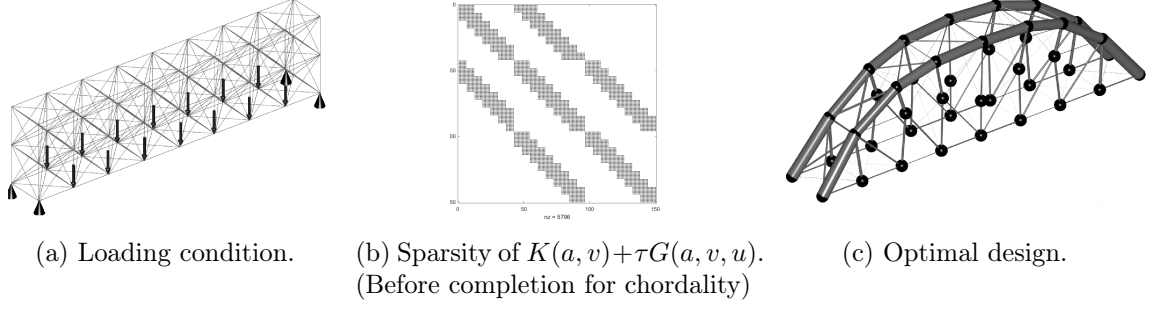


Figure 1: Example 2 – Loading condition, sparsity pattern of the resulting matrix, and optimal design.

- v is a decision variable for the nodal coordinates.
- u is a state variable for the nodal displacements.
- f is the external constant force.
- $\zeta > 0$ is a compliance bound.
- $\tau \geq 1$ is a loading factor.
- $l(v)$ is a vector of bar lengths such that

$$l_i(v) = \|v_i^{(2)} - v_i^{(1)}\|, \quad (21)$$

where $\|\cdot\|$ is Euclidean norm.

- $\mathcal{V}$ is a region defined by balls of radii r around the joints and is given by

$$\mathcal{V}_1 = \{v \in \mathbb{R}^{d_0 N} \mid \|v_j - \bar{v}_j\|^2 \leq r_j^2, j = 1, \dots, d_0\}. \quad (22)$$

- $K(a, v)$ is the global stiffness matrix, given by

$$K(a, v) = \sum_{i=1}^m a_i \frac{E}{l_i(v)} \gamma_i(v) \gamma_i^T(v), \quad (23)$$

where E is the Young's modulus of the material, and

$$\gamma_i(v) = \frac{1}{l_i(v)} (v_i^{(2)} - v_i^{(1)}, -(v_i^{(2)} - v_i^{(1)}))^T. \quad (24)$$

- $G(a, v, u_j)$ is the geometry stiffness matrix, given by

$$G(a, v, u_j) = \sum_{i=1}^m \frac{a_i E \gamma_i(v)^T u_j}{l_i^2(v)} (\delta_i(v) \delta_i(v)^T + \eta_i(v) \eta_i(v)^T), \quad (25)$$

where the vectors $\delta_i(v)$ and $\eta_i(v)$ must be computed such that $\gamma_i(v)$, $\delta_i(v)$, and $\eta_i(v)$ are mutually orthogonal.

The numerical statistics are presented in Table 2. The table also illustrates the efficiency of the decomposition, showing that the computational time per iteration was reduced by a factor of approximately 53. This improvement was achieved by completing for chordality and decomposing the 150×150 matrix in Figure 1b into 8 blocks, each of size between 30 and 36. Note that this problem was previously solved in [26], where the derivatives were computed manually

Table 2: Computational results and comparison of solution times with and without chordal decomposition for Example 2. A single 150×150 matrix was decomposed into 8 blocks of size between 30 and 36.

Without chordal decomposition			With chordal decomposition		
f_val	IPM iter	CPU (s)	f_val	IPM iter	CPU (s)
4.3368	86	3166	4.3373	64	44

Table 3: Computational results for Example 3.

Sizes (M, q)	IPM iter	CPU (s)
(12,000,6)	21	<1
(12,000,8)	19	3
(12,000,10)	46	13
(12,000,12)	27	13

and the chordal decomposition was performed in advance, not by the solver. In contrast, in this paper, the derivatives are obtained via automatic differentiation, and the decomposition is carried out using the built-in feature of IPNSDP. The complete code for this example is provided in `t_buckgeo.m`.

Example 3. This example presents a financial model that can be applied to classify failing and non-failing companies. The model is based on a Logit framework with a positive semidefinite quadratic discriminant function. A similar learning problem has been studied in [29]. The objective is to maximize the following function:

$$\begin{aligned} \max_Q \quad & \sum_{i=1}^M \left(y_i z(x_i) - \log \left(1 + e^{z(x_i)} \right) \right) \\ \text{subject to} \quad & Q \succeq 0, \end{aligned} \tag{26}$$

where, the parameters (variables) to be optimized are $a \in \mathbb{R}$, $b \in \mathbb{R}^q$, and $Q \in \mathbb{S}^q$. The function $z(x)$ is defined as

$$z(x) = a + b^T x + \frac{1}{2} x^T Q x$$

Here, $x_i \in \mathbb{R}^q$ represents the financial data for company i , and $y_i \in \{0, 1\}$ is a binary indicator ($y_i = 0$ for non-failure, $y_i = 1$ for failure). In practice, this problem involves a large number of learning data M and a small number of features q . Although the problem is nonlinear and potentially prone to numerical instability due to the exponential term in the objective function, the number of the optimization variables is relatively small, and the problem can typically be solved in a few seconds. However, it has been observed that for significantly large values of M , the setup phase in CasADi, can take minutes. For the numerical experiments, the entries of matrix M are generated from standard normal data adjusted using random average and spread vectors and then normalized between 0 and 1. See Table 3 for computational results. Note that for this example a 'separate' step strategy was used to determine the step length for the search directions. The complete code for this example is included in `f_d.m` in the distribution of the solver.

Example 4. This example is adapted from [20]. It involves a bilinear matrix inequality and is

Table 4: Computational results for Example 4.

Sizes (n, d, m)	IPM iter	CPU (s)
(50,25,50)	25	6
(50,50,50)	28	7
(100,50,100)	28	99
(100,100,100)	33	162

formulated as

$$\begin{aligned}
& \min_{x \in \mathbb{R}^n, \lambda \in \mathbb{R}} \lambda \\
& \text{s.t.} \quad -|c^\ell| \leq x^\ell \leq |c^\ell|, \quad \ell = 1, \dots, n, \\
& \quad B(x) = A_0 + \sum_{k=1}^n x_k A_k + \sum_{i=1}^d \sum_{j=1}^d x_i x_j K_{i,j} \preceq \lambda I_m,
\end{aligned} \tag{11.1}$$

where $A_k \in \mathbb{S}^m$ for $k = 0, \dots, n$, $K_{i,j} \in \mathbb{S}^m$ for $i, j = 1, \dots, d$, and $0 < d \leq n$. The bounds c^ℓ , $\ell = 1, \dots, n$, and the entries of the matrices A_k and $K_{i,j}$ are randomly drawn from the standard normal distribution. All matrices are approximately 20% dense; however, the aggregated matrix $B(x)$ is fully dense (100%), which makes the problem unsuitable for chordal decomposition, particularly in this setting. However, if $B(x)$ possesses a structured sparsity pattern, likely arising from specific applications, such problems could potentially be solved efficiently by IPNSDP. The numerical results are presented in Table 4. The implementation is provided in `bmi_1.m`, included in the solver distribution.

Example 5. In this example, a problem known by a Gaussian channel capacity problem is solved. This has been solved in [28, 29]. It is formulated as

$$\begin{aligned}
& \underset{x_i, t_i}{\text{minimize}} && \frac{1}{2} \sum_{i=1}^n \log(1 + t_i) \\
& \text{subject to} && \frac{1}{n} \sum_{i=1}^n x_i \leq 1 \\
& && x_i \geq 0, \quad t_i \geq 0, \quad i = 1, \dots, n \\
& && \begin{bmatrix} 1 - a_i t_i & \sqrt{r_i} \\ \sqrt{r_i} & a_i x_i + r_i \end{bmatrix} \succeq 0, \quad i = 1, \dots, n.
\end{aligned}$$

The problem features a nonlinear objective function and a large number n of linear matrix inequality constraints, each of dimension 2×2 . Due to the relatively small size of these matrix inequalities, the problem can be efficiently solved for larger instances compared to the other examples discussed in this report. For example, a problem involving 3200 such matrices was solved in under two minutes. For the results reported in Table 5, the parameters r_i and a_i , for $i = 1, \dots, n$, are randomly generated within the interval $[0, 1]$. The implementation code is provided in `g_c_cap.m`.

Remark 6. The IPNSDP software distribution includes additional examples, such as positive semidefinite programming extensions of selected test cases from the Hock and Schittkowski libraries [8] for testing nonlinear programming solvers and widely used in the NSDP literature [4, 29], the minimization of the minimum eigenvalue problem, and the primal and dual formulations of the well-known matrix completion problem.

Table 5: Computational results for Example 5.

Number of matrices n	IPM iter	CPU (s)
100	24	<1
200	27	<1
400	27	2
800	31	6
1600	33	20
3200	35	74

Remark 7. *In general, IPNSDP obtains solutions within a reasonable number of iterations when the problems exhibit convexity and when it is provided with feasible initial points. Although it includes features such as scaling and checks for linear constraint feasibility, it is still recommended that users supply a feasible initial point. Furthermore, for some problems, performance can be improved by customizing the algorithmic settings or the penalization strategy through its options.*

7 Conclusion

The paper introduces IPNSDP, an alternative MATLAB solver designed to address Nonlinear Semidefinite Programming (NSDP) problems using a second-order interior point method. IPNSDP aims to simplify the user experience by integrating with CASADi for automatic differentiation, thereby reducing the manual effort typically required for derivative computations involving matrix calculus.

For large-scale problems with structured sparse matrix constraints, IPNSDP leverages a chordal decomposition technique. This approach decomposes a single large positive semidefinite constraint into multiple smaller ones, significantly improving computational efficiency. To further improve the efficiency, the solver applies a clique merging strategy that combines submatrices with substantial overlap. This procedure is particularly beneficial for applications such as structural optimization, where these problem structures frequently occur.

Additionally, IPNSDP provides customization options for both the interior point algorithm and the decomposition procedures, enabling users to fine-tune the solver’s performance. Overall, IPNSDP offers a practical and user-friendly framework that combines advanced algorithmic strategies with automated tools to effectively solve a wide class of NSDPs.

References

- [1] Joel A E Andersson, Joris Gillis, Greg Horn, James B Rawlings, and Moritz Diehl. CasADi – A software framework for nonlinear optimization and optimal control. *Mathematical Programming Computation*, 11(1):1–36, 2019.
- [2] Santiago G. Constante F., Juan Camilo López, and Marcos J. Rider. Optimal reactive power dispatch with discrete controllers using a branch–and–bound algorithm: A semidefinite relaxation approach. *IEEE Transactions on Power Systems*, 36(5):4539–4550, 2021.
- [3] David Eppstein, Maarten Löffler, and Darren Strash. Listing all maximal cliques in sparse graphs in near-optimal time. In Otfried Cheong, Kyung-Yong Chwa, and Kunsoo Park, editors, *Algorithms and Computation*, pages 403–414, Berlin, Heidelberg, 2010. Springer Berlin Heidelberg.
- [4] Jan Fiala, Michal Kočvara, and Michael Stingl. *PENLAB: a MATLAB solver for nonlinear semidefinite optimization*, 2013.

- [5] Mitsuhiro Fukuda, Masakazu Kojima, Kazuo Murota, and Kazuhide Nakata. Exploiting sparsity in semidefinite programming via matrix completion i: General framework. *SIAM Journal on Optimization*, 11(3):647–674, 2001.
- [6] Michael Garstka, Mark Cannon, and Paul Goulart. A clique graph based merging strategy for decomposable SDPs. *IFAC–PapersOnLine*, 53(2):7355–7361, 2020.
- [7] Nicholas J. Higham. Computing the nearest correlation matrix—a problem from finance. *IMA Journal of Numerical Analysis*, 22(3):329–343, 07 2002.
- [8] W. Hock and K. Schittkowski. *Test Examples for Nonlinear Programming Codes*. Springer-Verlag, Berlin, 1981.
- [9] Sunyoung Kim, Masakazu Kojima, Martin Mevissen, and Makoto Yamashita. Exploiting sparsity in linear and nonlinear matrix inequalities via positive semidefinite matrix completion. *Math. Program.*, 129(1):33–68, 2011.
- [10] Michal Kocvara, F Leibfritz, Michael Stingl, and D Henrion. A nonlinear sdp algorithm for static output feedback problems in compleib. *Proceedings of the 16th IFAC World Congress on Automatic Control, Prague*, 16, 01 2005.
- [11] Hiroshi Konno, Naoya Kawada, and Dai Wu. Estimation of failure probability using semi-definite logit model. *Computational Management Science*, 1(1):59–73, 2003.
- [12] M. Kočvara. On the modelling and solving of the truss design problem with global stability constraints. *Structural and Multidisciplinary Optimization*, 23(3):189–203, Apr 2002.
- [13] Michal Kočvara. Decomposition of arrow type positive semidefinite matrices with application to topology optimization. *Mathematical Programming*, 190:105–134, 06 2020.
- [14] Michal Kočvara and Michael Stingl. Pennon: A code for convex nonlinear and semidefinite programming. *Optimization Methods and Software*, 18(3):317–333, 2003.
- [15] J. Lofberg. Yalmip : a toolbox for modeling and optimization in matlab. In *2004 IEEE International Conference on Robotics and Automation (IEEE Cat. No.04CH37508)*, pages 284–289, 2004.
- [16] MOSEK ApS, Copenhagen, Denmark. *MOSEK Optimization Toolbox for MATLAB Manual*, 2024.
- [17] Y. E. Nesterov and M. J. Todd. Self-scaled barriers and interior-point methods for convex programming. *Mathematics of Operations Research*, 22(1):1–42, 1997.
- [18] Y. E. Nesterov and M. J. Todd. Primal-dual interior-point methods for self-scaled cones. *SIAM Journal on Optimization*, 8(2):324–364, 1998.
- [19] Pablo A Parrilo. Semidefinite programming relaxations for semialgebraic problems. *Mathematical Programming*, 96(2):293–320, 2003.
- [20] M. Stingl. *On the Solution of Nonlinear Semidefinite Programs by Augmented Lagrangian Method*. PhD thesis, Institute of Applied Mathematics II, Friedrich-Alexander University of Erlangen-Nuremberg, 2006.
- [21] Jos F. Sturm. Using sedumi 1.02, a matlab toolbox for optimization over symmetric cones. *Optimization Methods and Software*, 11(1-4):625–653, 1999.

- [22] Kim-Chuan Toh, Michael J. Todd, and Reha H. Tütüncü. SDPT3 — a Matlab software package for semidefinite programming, version 1.3. *Optimization Methods and Software*, 11(1–4):545–581, 1999.
- [23] Lieven Vandenbergh and Martin S. Andersen. Chordal graphs and semidefinite optimization. *Foundations and Trends® in Optimization*, 1(4):241–433, 2015.
- [24] Alemseged Weldeyesus, Jacek Gondzio, Linwei He, Matthew Gilbert, Paul Shepherd, and Andrew Tyas. Truss geometry and topology optimization with global stability constraints. *Structural and Multidisciplinary Optimization*, 62:1721–1737, 2020.
- [25] Alemseged Weldeyesus and Mathias Stolpe. A primal-dual interior point method for large-scale free material optimization. *Computational Optimization and Applications*, 61(2):409–435, 2015.
- [26] Alemseged Gebrehiwot Weldeyesus, Jacek Gondzio, and Miguel F. Anjos. On the scalability of truss geometry and topology optimization with global stability constraints via chordal decomposition. *Structural and Multidisciplinary Optimization*, 68(1):7, 2024.
- [27] Stephen J. Wright. *Primal-Dual Interior-Point Methods*. SIAM, Philadelphia, 1997.
- [28] Yuya Yamakawa and Takayuki Okuno. A stabilized sequential quadratic semidefinite programming method for degenerate nonlinear semidefinite programs. *Computational Optimization and Applications*, 83(3), 2022.
- [29] H. Yamashita, H. Yabe, and K. Harada. A primal-dual interior point method for nonlinear semidefinite programming. *Mathematical Programming*, 135(A):89–121, 2012.
- [30] Hiroshi Yamashita and Hiroshi Yabe. A survey of numerical methods for nonlinear semidefinite programming. *Journal of the Operations Research Society of Japan*, 58(1):24–60, 2015.
- [31] Mihalis Yannakakis. Computing the minimum fill-in is np-complete. *SIAM Journal on Algebraic Discrete Methods*, 2(1):77–79, 1981.
- [32] Yang Zheng, Giovanni Fantuzzi, and Antonis Papachristodoulou. Chordal and factor-width decompositions for scalable semidefinite and polynomial optimization. *Annual Reviews in Control*, 52:243–279, 2021.

A Complete IPNSDP Problem Setup Template

```

1 function ipnsdp_input_template()
2 %IPNSDP_INPUT_TEMPLATE Example template for defining an NSDP in ipnsdp.
3 %
4 % This script sets up a generic Nonlinear Semidefinite Program (NSDP)
5 % of the form:
6 %
7 % minimize f(X, u)
8 % subject to
9 %     sum_{k=1}^{m_X} <A11_i^{(k)}, X_k> + A12_i * u = b1_i, i = 1,...,n1
10 %     sum_{k=1}^{m_X} <A21_j^{(k)}, X_k> + A22_j * u <= b2_j, j = 1,...,n2
11 %     c1(X, u) = 0
12 %     c2(X, u) <= 0
13 %     A_s(X, u) >= 0 (PSD), s = 1,...,n_psd
14 %     u_lower <= u <= u_upper
15 %     rho_k_lower * I_{d_k} <= X_k <= rho_k_upper * I_{d_k}, k = 1,...,m_X
16 %

```

```

17 % where
18 %   -  $X_k$  in  $S^{d_k}$ ,  $k = 1, \dots, m_X$ , are symmetric matrix variables
19 %   -  $u$  in  $R^n$  is a vector of scalar variables
20 %   -  $\langle A, X \rangle = \text{trace}(A' * X)$  denotes the Frobenius inner product
21 %
22 %% Problem Definition
23 % All fields are listed with empty values. Comments explain intended usage.
24
25 prob.name      = '';           % Name of the problem (optional)
26 prob.problem_data = struct(); % Struct to store user data (optional)
27
28 % Matrix variables X
29 prob.nX        = [];           %  $m_X$ : number of matrix variables  $X_k$ .
30 %                               % Set to 0 if none.
31 prob.dimX      = [];           %  $[d_1, \dots, d_{m_X}]$ : sizes  $d_k$  of  $X_k$ .
32 %                               % Set to [] if none.
33
34 % Bounds on Matrix variables X
35 prob.lbX       = [];           %  $[\text{rho}_{1\_lower}, \dots, \text{rho}_{m\_X\_lower}]$ 
36 %                               % Set to [] if all unbounded.
37 %                               % Use -Inf for selectively unbounded entries.
38 prob.ubX       = [];           %  $[\text{rho}_{1\_upper}, \dots, \text{rho}_{m\_X\_upper}]$ 
39 %                               % Set to [] if all unbounded.
40 %                               % Use +Inf for selectively unbounded entries.
41
42 % Scalar variables u
43 prob.nu        = [];           %  $n$ : dimension of vector  $u$ .
44 %                               % Set to 0 if none.
45
46 % Bounds on scalar variables u
47 prob.lbu       = [];           %  $[u_{lower\_1}; \dots; u_{lower\_n}]$ 
48 %                               % Set to [] if all unbounded.
49 %                               % Use -Inf for selectively unbounded entries.
50 prob.ubu       = [];           %  $[u_{upper\_1}; \dots; u_{upper\_n}]$ 
51 %                               % Set to [] if all unbounded.
52 %                               % Use +Inf for selectively unbounded entries.
53
54 % nPSD constraints
55 prob.nPSDcon = [];           %  $n\_psd$ : number of PSD constraints  $A_s(X, u) \succeq 0$ .
56 %                               % Set to 0 if none.
57
58 % Linear constraints
59 % Note: You may embed these into nonlinear constraints if easier.
60 prob.A11 = [];           %  $[\text{vec}(A11\_1\_1)^T, \dots, \text{vec}(A11\_1\_mX)^T;$ 
61 %                               % ...;
62 %                               %  $\text{vec}(A11\_n1\_1)^T, \dots, \text{vec}(A11\_n1\_mX)^T]$ 
63 %                               % Coefficient matrix for  $[\text{vec}(X_1); \dots; \text{vec}(X_{mX})]$ 
64 %                               % in linear equality constraints. [] if none.
65 prob.A12 = [];           % Coefficient matrix for scalar variables  $u$  in
66 %                               % linear equalities. [] if none.
67 prob.b1 = [];           % RHS vector of linear equality constraints.
68 %                               % [] if none.
69 prob.A21 = [];           %  $[\text{vec}(A21\_1\_1)^T, \dots, \text{vec}(A21\_1\_mX)^T;$ 
70 %                               % ...;
71 %                               %  $\text{vec}(A21\_n2\_1)^T, \dots, \text{vec}(A21\_n2\_mX)^T]$ 
72 %                               % Coefficient matrix for  $[\text{vec}(X_1); \dots; \text{vec}(X_{mX})]$ 
73 %                               % in linear inequality constraints. [] if none.
74 prob.A22 = [];           % Coefficient matrix for scalar variables  $u$  in

```

```

75         % linear inequalities. [] if none.
76 prob.b2 = [];      % RHS vector of linear inequality constraints.
77                 % [] if none.
78
79 % Objective function
80 prob.obj = @(x) objective(x, prob);
81                 % User-defined objective function
82 % Nonlinear and PSD constraints
83 prob.nlcon = @(x) nonlinear_constraint(x, prob);
84                 % User-defined nonlinear constraints
85                 % Must return [eq, ineq, psd]:
86                 % eq  = nonlinear equalities (vector)
87                 % ineq = nonlinear inequalities (vector)
88                 % psd = cell array of PSD matrices
89
90 % Initial point (optional)
91 prob.x0.X = {};    % Cell array of initial guesses for X_k.
92                 % Example: prob.x0.X{1} = eye(d_1), ...,
93                 %         prob.x0.X{m_X} = eye(d_m_X)
94                 % Defaults are used if not specified.
95 prob.x0.u = [];    % Column vector initial guess for u.
96                 % Default is zero if not specified.
97
98 %% Algorithmic Options
99 % Default solver parameters (modifiable by the user)
100
101 prob.options.scale      = 'on';  % 'on'/'off': Enable/Disable scaling
102 prob.options.opt_tol    = 1e-6;  % Optimality tolerance
103 prob.options.feas_tol   = 1e-7;  % Feasibility tolerance
104 prob.options.max_iter   = 100;   % Maximum number of iterations
105 prob.options.min_mu     = 1e-9;  % Minimum barrier parameter
106 prob.options.sigma      = 0.4;   % Centrality parameter
107 prob.options.tau        = 0.9;   % Backtracking step size
108 prob.options.gamma      = 0.99;  % Step length scaling
109 prob.options.stepMode   = 'joint'; % 'joint'/'separate': Step strategy
110 prob.options.chord_decomp = 'yes'; % 'yes'/'no': Chordal decomposition
111 prob.options.clique_merge = 'yes'; % 'yes'/'no': Clique merging
112 prob.options.density     = 0.7;  % Density threshold for decomposition
113 prob.options.min_block_size = 30; % Minimum block size for decomposition
114 prob.options.rel_block_size = 0.6; % Relative block size for decomposition
115 prob.options.linconCheck = 'yes'; % 'yes'/'no': LinConst dependency test
116 prob.options.nonlinconCheckN = 'yes'; % 'yes'/'no': NonlinConst dependency test
117                                     (Numeric)
118 prob.options.nonlinconCheckS = 'no'; % 'yes'/'no': NonlinConst dependency test
119                                     (Symbolic)
120 prob.options.penlpMode    = 'off'; % 'off'/'on': Penalization
121 prob.options.penlp        = 0;     % Penalty parameter (0 = off)
122 prob.options.penldeg      = 2;     % 1 or 2: Penalty degree
123
124 %% Solve the problem
125 [x_sol, info] = ipnsdp_solve(prob);
126
127 %% Results (saved to ipnsdp_sol.mat)
128 x_sol.X      % Cell array of solution X_k matrices or slacks
129 x_sol.u      % Optimized scalar variables vector u
130 x_sol.Z      % Dual vars for PSD constraints X_k and A_s(X,u)
131 x_sol.lambda1 % Dual variables for linear equalities
132 x_sol.lambda2 % Dual variables for linear inequalities

```

```

131 x_sol.theta1      % Dual variables for nonlinear equalities
132 x_sol.theta2      % Dual variables for nonlinear inequalities
133 x_sol.slinear      % Slack variables for linear inequalities
134 x_sol.snonlinear   % Slack variables for nonlinear inequalities
135
136 info.obj_evals      % objective function evaluations
137 info.grad_evals     % gradient evaluations
138 info.nonlincon_evals % nonlinear constraint evaluations
139 info.nonlincon_jacobi_evals % nonlinear constraint Jacobian evals
140 info.Lag_hessian_evals % Lagrangian Hessian evaluations
141 info.CPU            % CPU time used (seconds)
142 info.iter           % Number of iterations performed
143 info.obj_value       % Final objective function value
144 info.opt_tol         % Final optimality tolerance
145 info.primal_feasibility % Final primal feasibility measure
146 info.dual_feasibility % Final dual feasibility measure
147 info.complementarity % Final complementarity gap
148 info.obj_val_hist    % History of objective values
149 info.kkt_optimality  % History of KKT conditions
150 info.exit            % Solver exit message
151
152 end
153
154 %% Objective Function
155 function f_val = objective(x, prob)
156 % Computes the objective function value.
157 %
158 % Inputs:
159 %   x : struct with fields:
160 %       x.X : cell array of matrix variables
161 %       x.u : vector of scalar variables
162 %   prob: problem-specific data struct
163 %
164 % Output:
165 %   f_val: scalar objective value
166 %
167 % ... (user-defined implementation)
168 end
169
170 %% Nonlinear and PSD Constraints
171 function [eq, ineq, psd] = nonlinear_constraint(x, prob)
172 % Computes nonlinear equality, inequality, and PSD constraints.
173 %
174 % Inputs:
175 %   x.X : cell array {X_1, ..., X_mX}, each X_k in S^{d_k}
176 %   x.u : vector of scalar variables u in R^n
177 %   prob: problem data struct
178 %
179 % Outputs:
180 %   eq : nonlinear equality constraints vector (if none set to [])
181 %   ineq : nonlinear inequality constraints vector (if none set to [])
182 %   psd : psd{1} = A_1(X, u), ..., psd{n_psd} = A_n_psd(X, u)
183 %         cell array of PSD constraint matrices (if none set to {})
184 %
185 % ... (user-defined implementation)
186 end

```

Listing 1: ipnsdp_user_template.m – user input template for IPNSDP